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SUPERSEDING
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MILITARY SPECIFICATION

CONTROL UNITS, REMOTE SWITCHING, C-10315/U and C-10979/U

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, production, and testing requirements for Remote Switching Control Units C-10315/U and C-10979/U for use with shipboard secure voice communications systems.

1.2 Classification. The control units shall be of the following types, as specified (see 6.2.1):

Type I - For use with surface ships. Provides programming capability for nine telephone sets and five channels.
Type II - For use with smallcraft and submarines. Provides programming capability for three telephone sets and three channels.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue listed in the Department of Defense Index of Specifications and Standards (DoDISS) and its supplements, form a part of this document to the extent specified herein. The date of the applicable DoDISS and supplements thereto shall be as specified in the solicitation.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment And Systems, Requirements For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore, General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of

STANDARDS

FEDERAL

FED-STD-595	Color

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-965	Parts Control Program
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)

DRAWINGS

MILITARY

NAVAL ELECTRONIC SYSTEMS COMMAND

DL 76E5N100	Data List, Remote Switching Control C-10315/U
DL 80E2N100	Data List, Remote Switching Control C-10979/U

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The control units shall conform to the requirements of MIL-E-16400 to the extent specified herein.

3.2 Classification of requirements. The requirements for the control units are classified as follows:

Requirement	Paragraph
First article	3.3
Parts, materials, and processes	3.4
Design and construction	3.5
Reliability	3.6
Maintainability	3.7
Performance characteristics	3.8
Environmental requirements	3.9
Electromagnetic interference and electromagnetic compatibility	3.10
Physical requirements	3.11
Color and finish	3.12
Workmanship	3.13

3.3 First article. When specified (see 6.2.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

3.4 Parts, materials, and processes. Parts, materials, and processes used in production of the control units shall conform to the requirements of MIL-E-16400.

3.4.1 Parts control. The parts to be incorporated in the control units shall be controlled in accordance with MIL-STD-965, Procedure I.

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3.5 Design and construction. Design and construction of the control units shall be in conformance with the requirements of MIL-E-16400 and drawing (DWG) DL 76E5N100 for Type I control units and DWG DL 80E2N100 for Type II control units to the extent specified herein. The control units shall be manufactured in accordance with the drawings and specifications listed in DWG DL 76E5N100 and related lists for Type I control units and in DWG DL 80E2N100 and related lists for Type II control units, except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this specification.

3.5.1 Control unit composition. The control units shall consist of the following:

a. Control panel
b. Audio amplifiers
c. Power supply

3.5.1.1 Control panel. The control panel shall contain the necessary controls, indicators, and switches required to conform to the requirements specified herein. The panel shall contain a matrix type array of:

a. Type I - 54 illuminated pushbutton switches
b. Type II - 12 illuminated pushbutton switches

3.5.1.2 Audio amplifier. Each amplifier card shall provide for amplification in the transmit and the receive mode. Characteristics of the transmit and receive amplifiers shall be as specified in 3.8.8.1 and 3.8.8.2.

3.5.1.2.1 Type I. Five plug-in, solid-state audio amplifier cards shall be provided, one for each channel.

3.5.1.2.2 Type II. Three plug-in, solid-state audio amplifier cards shall be provided, one for each channel.

3.5.1.3 Power supply. A 24 volts direct current (VDC) regulated power supply shall be provided to supply power to external equipment connected to the control units (see 3.8.6).

3.5.2 Electrical power requirements. Type I and Type II control units shall conform to the electrical requirements specified in 3.5.2.1 through 3.5.2.6.

3.5.2.1 Primary power supply line voltage. The control units shall be designed to operate from shipboard alternating current (AC) power as specified in Section 300 of DoD-STD-1399, nominal. The control units shall also be capable of operation from an external +24 VDC ±5 percent source (not specified in this specification).

3.5.2.2 Transient voltage. The control units shall conform to the transient voltage requirements of MIL-E-16400.

3.5.2.3 Electrical bonding. The control units shall conform to the electrical bonding requirements of MIL-E-16400.

3.5.2.4 Internal cabling. The control units shall conform to the internal cabling requirements of MIL-E-16400.

3.5.2.5 Wire terminations. The control units shall conform to the wire termination requirements of MIL-E-16400.

3.5.2.6 Blown fuse indicators. Blown fuse indicators are not required.

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3.5.3 Front panel layout and arrangement of operating controls. Front panel layout and arrangement of operating controls shall provide the following:

a. Type I:
   1. Pushbutton switches for interconnection of up to nine remote telephone sets to up to five secure radio channels. The unit shall be capable of being cascaded with other units by transfer switches, thus providing for expanded switching capability.
   2. A locking mechanism to lock the front panel switches in the selected positions.

b. Type II:
   1. Pushbutton switches for interconnection of up to three remote telephone sets to up to three secure radio channels.
   2. A locking mechanism to lock the front panel switches in the selected positions.

3.5.4 Cable entrances. Cable entrances shall be provided on top, botton, and two sides of case, conforming to the cable entrances specified in DWG DL 76E5N100 for Type I control units top and two sides of the case, conforming to the cable entrances specified in DWG DL 80E2N100 for Type II control units.

3.5.5 Safety. The control units shall be in accordance with the requirements of the Safety criteria paragraph of MIL-E-16400.

3.6 Reliability.

3.6.1 Quantitative reliability requirement. The specified upper test mean-time-between-failures (MTBF) (θ_0 as defined in MIL-STD-781) for Type I and Type II control units shall be 1500 hours. The lower test MTBF (θ_1 as defined in MIL-STD-781) shall be 500 hours.

3.7 Maintainability.

3.7.1 Quantitative maintainability requirements. Repairs for Type I and Type II control units shall be accomplished by replacement of lowest subassemblies (modules or printed wiring assemblies) or chassis-mounted parts. The control unit mean-corrective-maintenance-time (M_{ct}) shall be as follows:

a. Type I. Not exceeding 90 minutes
b. Type II. Not exceeding 30 minutes

3.8 Performance characteristics. The control units shall conform to the performance characteristics specified in 3.8.1 through 3.8.8.2.

3.8.1 Switching requirements. All switch contacts shall conform to the following characteristics:

a. Resistance (switch closed). 5 ohms maximum
b. Resistance (switch open). 100 kilohms (kohms) minimum
c. Current. One ampere maximum, noninductive

3.8.2 Switching function characteristics.

3.8.2.1 Type I. The pushbutton with which the switching function is performed shall be arranged in a rectangular matrix consisting of six rows and nine columns. Rows one through five, counting from the top down, shall be associated with secure radio sets one through five, respectively. Row six shall be associated with the transfer and release functions. Columns one through nine shall be associated with remote telephone sets one through nine, respectively. The depressing of any one switch in rows one through five in a vertical column shall connect the radio associated with that respective row with the telephone associated with the selected

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column. Depressing a pushbutton in row six shall release any previously depressed pushbutton in that respective column and shall transfer the remote telephone set connected to that column to the cascaded switching unit.

3.8.2.2 Type II. The pushbuttons with which the switching function is performed shall be arranged in a rectangular matrix consisting of four rows and three columns. Rows two through four, counting from top down, shall be associated with secure radio sets one through three, respectively. The depressing of any one switch in rows two through four in a vertical column shall connect the radio associated with that respective row with the telephone associated with the selected column. Depressing a pushbutton in row one shall release any depressed pushbutton in that respective column.

3.8.3 Interlock. Type I and Type II control units shall be designed so that only one pushbutton in a vertical column can be depressed at any time.

3.8.4 Input interface characteristics. The input interface for both Type I and Type II control units shall be capable of operating with the input signals as specified in 3.8.4.1 and 3.8.4.2, and providing the signals in 3.8.4.3 through 3.8.4.5.

3.8.4.1 Transmit audio. The telephone set shall provide an audio signal with the following characteristics:

a. Level from telephone. -6 decibels referred to one milliwatt (dBm) average, +14 dBm peak
b. Frequency characteristic of telephone. Flat within 8 decibels (dB) peak-to-peak from 200 hertz (Hz) to 4 kilohertz (kHz)
c. Transmit input impedance. 600 ohms ±10 percent at 1 kHz balanced

3.8.4.2 Receive audio. The radio set shall provide an audio signal with the following characteristics:

a. Level from radio. 0 dBm average, +20 dBm peak
b. Frequency characteristic of radio. Flat within 6 dB peak-to-peak from 200 Hz to 4 kHz
c. Receive input impedance. 600 ohms ±10 percent at 1 kHz, balanced

3.8.4.3 Disconnect signal. A disconnect signal shall be provided at the telephone interface when no connection to the telephone is made on the matrix switches. The interface shall conform to the following characteristics:

a. Disconnect. +24 volts (V) ±10 percent with series resistance of 150 ohms +20 percent
b. Connect. Ground with a series resistance of 150 ohms ±20 percent

3.8.4.4 Plus 24 VDC. Plus 24 VDC shall be supplied to each telephone, unswitched, and derived from the power supply specified in 3.5.2.1 or 3.8.6.

3.8.4.5 Signal ground. Each telephone interface shall have a terminal for signal ground.

3.8.5 Output interface characteristics. The output interface characteristics for both Type I and Type II control units shall be as specified in 3.8.5.1 through 3.8.5.3.

3.8.5.1 Transmit audio. The transmit audio to the radio shall conform to the following characteristics:

a. Level to the radio. -6 dBm average, +14 dBm peak
b. Frequency response. Frequency response shall be flat within 6 dB peak-to-peak from 200 Hz to 4 kHz reference to the input signal with a minimum rolloff of -20 dB per decade.
c. Transmit output impedance. 600 ohms ±10 percent at 1 kHz, balanced
d. Distortion. Three percent maximum

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e. Noise. The signal-to-noise ratio shall be greater than 40 dB referenced to a 0 dBm signal level. Noise shall be considered hum and any other extraneous signal. All inputs and outputs including the one under test shall be terminated into 600 ohms ±10 percent
f. Gain. The gain shall be unity +0 dB, -3 dB measured at 1 kHz.

3.8.5.2 Receive audio. The receive audio to the telephone shall conform to the following characteristics:

a. Level to the telephone. 0 dBm average, +20 dBm peak
b. Frequency response. With a load of 30 ohms to 600 ohms the frequency response shall be flat within 7 dB peak-to-peak from 200 Hz to 4 kHz reference to the input signal with a minimum rolloff of -20 dB per decade.
c. Receive output impedance. 30 ohms maximum, balanced
d. Distortion. Five percent maximum
e. Noise. The signal-to-noise ratio shall be greater than 40 dB referenced to a 0 dBm input signal level. Noise shall be considered hum and any other extraneous signal noise. All audio inputs and outputs including the one under test shall be terminated into 600 ohms ±10 percent
f. Gain. The gain shall be unity +0 dB, -6 dB measured at 1 kHz.

3.8.5.3 Plus 24 VDC. Plus 24 VDC shall be supplied to each secure radio system unswitched, and derived as specified in 3.5.2.1 or 3.8.6.

3.8.6 Internal power supply. Both Type I and Type II control units shall contain a 24 VDC regulated power supply. The power supply shall be a plug-in type, and shall be removable from the front of the control unit.

3.8.6.1 Short circuit protection. Short circuit overload protection shall be provided so that a direct short on the direct current (DC) output will not damage the power supply. Such a provision shall allow resumption of operation immediately following removal of the overload.

3.8.6.2 Characteristics. The power supply shall conform to the following characteristics:

a. Voltage. +24 VDC ±5 percent
b. Current. 3.5 amperes maximum (Type I)
1 ampere maximum (Type II)
c. Ripple. 10 millivolts (mV) root-mean-square (rms) maximum at full load
d. Regulation. 2 percent (minimum load to full load) at the telephone interface

3.8.6.3 Switching capability (Type I only). A switching capability on the power supply shall be provided to permit selection of +24 VDC power from either the internal power supply or an external +24 VDC source.

3.8.7 Crosstalk. For Type I and Type II control units, there shall be a minimum of 80 dB isolation over the frequency range 300 Hz to 3000 Hz between any transmit inputs and receive circuits within the control units.

3.8.8 Audio amplifier card. The characteristics for all amplifier cards for Type I and Type II control units shall be as specified in 3.8.8.1 and 3.8.8.2.

3.8.8.1 Transmit mode. The transmit audio shall conform to the following characteristics:

. Input level. -26 dBm average, -6 dBm peak
Output level. -6 dBm average, +14 dBm peak
b. Frequency response. Frequency response shall be flat within 6 dB peak-to-peak from 200 Hz to 4 kHz reference to the input signal with a minimum rolloff of -20 dB per decade
c. Input impedance. 50 ohms ±10 percent at 1 kHz balanced
Output impedance. 600 ohms ±10 percent at 1 kHz balanced
d. Distortion. Two percent maximum

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e. Noise. The signal-to-noise ratio shall be greater than 40 dB. Noise shall be considered hum and any other extraneous signal noise. All audio inputs and outputs including the one under test shall be terminated into 600 ohms ±10 percent.
f. Gain. 20 dB, +0 dB, -3 dB measured at 1 kHz

3.8.8.2 Receive mode. The receive audio shall conform to the following characteristics:

a. Input level. 0 dBm average, +14 dBm peak
Output level. 0 dBm average, +14 dBm peak
b. Frequency response. With a load of 30 ohms to 600 ohms the frequency response shall be flat within 7 dB peak-to-peak from 200 Hz to 4 kHz reference to the input signal with a minimum rolloff of -20 dB per decade
c. Input impedance. 600 ohms ±10 percent at 1 kHz balanced
Output impedance. 30 ohms maximum, balanced
d. Distortion. Two percent maximum
e. Noise. The signal-to-noise ratio shall be greater than 40 dB. Noise shall be considered hum and any other extraneous signal. All inputs and outputs including the one under test shall be terminated into 600 ohms ±10 percent
f. Gain. The gain shall be unity +0 dB, -6 dB measured at 1 kHz

3.8.9 Telephone pads. A telephone resistor pad shall be provided for each channel in both Type I and Type II control units. These resistors shall provide a means of matching the 50-ohm input impedance of the transmit side of the audio amplifier card (see 3.8.8) to the incoming 600-ohm balanced line (see 3.8.4.1).

3.9 Environmental requirements. The environmental requirements for Type I and Type II control units shall be as specified in 3.9.1 through 3.9.4.

3.9.1 Temperature. Temperature conditions shall be in accordance with Range 4 of MIL-E-16400, except the nonoperating temperature shall be -55° Celsius (C) to +75° C.

3.9.2 Humidity. The control units shall conform to the humidity requirements of MIL-E-16400.

3.9.3 Shock. Shock requirements shall be in accordance with the requirements for Grade A, Class I, Type A lightweight equipment as specified in MIL-S-901.

3.9.4 Vibration. Vibration requirements shall be in accordance with MIL-E-16400.

3.10 Electromagnetic interference (EMI) and electromagnetic compatibility (EMC). Type I and Type II control units shall conform to the limit requirements for Class A4 and Class A5 equipment in accordance with MIL-STD-461 for the following tests only: CE01, CE03, CE07, CS01, CS06, CS09, RE02, RS01, and RS03. The equipment shall be self-compatible and compatible with co-located equipment normally working in its intended installation (see 4.5.4).

3.11 Physical requirements.

3.11.1 Size. The control units shall conform to the size as specified in 3.11.1.1 and 3.11.1.2.

3.11.1.1 Type I. Type I control units shall be capable of both standard 482.6 millimeters (mm) (19 inches (in)) rack mounting and table mounting and shall not exceed 304.8 mm (12 in) in depth and 355.6 mm (14 in) in height. Mounting brackets for rack mounting shall be furnished with each control unit.

3.11.1.2 Type II. Type II control unit dimensions shall not exceed 304.8 mm (12 in) in height, 228.6 mm (9 in) in width and 304.8 mm (12 in) in depth. The control unit shall be capable of bulkhead mounting with brackets and table mounting.

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3.11.2 Weight. The control units shall conform to the following weight requirements:

a. Type I. 40.82 kilograms (kg) (90 pounds (lbs)) maximum
b. Type II. 21.77 kg (48 lbs) maximum

3.11.3 Mounting dimensions. Mounting dimensions shall conform to the dimensions specified in DWG DL 76E5N100 for Type I control units and DWG DL 80E2N100 for Type II control units.

3.12 Color and finish. Color and finish shall be in accordance with the requirements of DWG DL 76E5N100 for Type I control units and DWG DL 80E2N100 for Type II control units. The finish shall be red enamel, No. 11136 of FED-STD-595.

3.13 Workmanship.

3.13.1 General workmanship. The workmanship and soldering shall conform to the Workmanship and Soldering paragraphs of MIL-E-16400.

3.13.2 Workmanship screen. All control units produced in accordance with this specification shall withstand a defect detection vibration screen of random type vibration at $0.04g^2/Hz$ ± 3 dB from 80 Hz to 3.50 Hz and temperature cycling as specified in 4.5.5.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and sevices conform to the prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection:
    1. Production inspection (Group A) (see 4.4.1)
    2. Production control inspection (Group B) (see 4.4.2)
    3. Environmental inspection (Group C) (see 4.4.3)
c Reliability test (see 4.6)
  Maintainability demonstration (see 4.7)
e Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE I.

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4.4 Quality conformance inspection.

4.4.1 Production inspection (Group A). Production inspection shall be made on every control unit offered for delivery. The inspection shall comprise such examination and testing which will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the examinations and tests shown in Group A of TABLE I. Workmanship shall be the first inspection performed.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis in conformance with the inspection procedures of MIL-STD-105 to an acceptable quality level (AQL) of 6.5 percent using the special inspection level S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on control units that have been subjected to and have passed production inspection. Production control inspection shall consist of the examination and functional and performance tests throughout the entire range of operation; detect any deterioration of the design by wear of dies, molds, and jigs, by variations in parts; deviations in the processing temperature rise in operation and ability of control units to withstand this heat; tests of efficiency; and tests of the control unit performance. These tests shall be performed on the complete control unit as offered for delivery. Production control inspection shall include the examinations and tests shown in Group B of TABLE I.

4.4.3 Environmental inspection (Group C). Environmental inspection shall encompass tests of control set durability, simulated service tests; test of the effects of environment (such as extremes of temperature and humidity); and vibration. Environmental inspection shall include the examination and tests shown in Group C of TABLE I. Environmental inspection shall be performed on a sampling basis on control units which have been subjected to and have passed production inspection (see 4.4.3.1).

4.4.3.1 Environmental inspection samples. Samples for the environmental inspection shall be selected, without regard to their quality, in accordance with the following:

a. One control unit from first month's production
b. One control unit from each successive 100 control units, or fraction thereof, produced

TABLE I. Examinations and test.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination	3.5	4.5.1	X			
Weight	3.11.2	4.5.1	X		X	
Mounting dimensions	3.11.3		X	X		
Size	3.11.1	4.5.1	X	X		
Parts and materials	3.4	4.5.1	X	X		
Color and finish	3.12	4.5.1	X	X		
Marking	3.4	4.5.1	X	X		
Workmanship						
General workmanship	3.13.1	4.5.5.1	X	X		
Workmanship screen	3.13.2	4.5.5.2		X		
Pre-performance		4.5.1	X			X
Performance	3.8	4.5.2	X	X		

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TABLE I. Examinations and test. (Continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Input and output interface characteristics	3.8.4, 3.8.5	4.5.2	X	X		
Switching requirements	3.8.1	4.5.2	X	X		
Switching function characteristics	3.8.2	4.5.2	X	X		
Interlock	3.8.3	4.5.2	X	X		
Power supply characteristics	3.8.6.2	4.5.2	X	X		
Audio amplifier card	3.8.8	4.5.2	X	X		
Switching capability	3.8.6.3	4.5.2	X	X		
Pad-impedance match	3.8.9	4.5.2	X	X		
Crosstalk	3.8.7	4.5.7	X		X	
Temperature range						
Low temperature	3.9.1	4.5.3	X			
High temperature	3.9.1	4.5.3	X			
Humidity	3.9.2	4.5.1	X			X
Shock	3.9.3	4.5.1	X			
Vibration	3.9.3	4.5.1	X			X
Steady-state voltage and frequency	3.5.2	4.5.1	X		X	
Transient voltage	3.5.2	4.5.1	X			
Transient frequency	3.5.2	4.5.1	X			
Spike voltage	3.5.2	4.5.1	X			
Power interruption	3.5.2	4.5.1	X		X	
Power and power factor	3.5.2	4.5.1	X			
Leakage current	3.5.5	4.5.6	X			X
Electromagnetic interference	3.10	4.5.4	X			
Cable entrance	3.5.4		X			

4.5 Test methods.

4.5.1 Examinations and tests. Examinations and tests shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.5.2 through 4.5.7.

4.5.2 Performance test. Type I and Type II control units shall be energized and subjected to a performance test to ensure proper functioning and operation at an adequate number of points in the required range to demonstrate conformance to specification requirements, including safety. The performance test shall include, but not be limited to, verification of compliance with the requirements shown in TABLE I.

4.5.3 Temperature tests. Temperature tests shall be conducted to determine conformance with the requirements specified in 3.9.1.

4.5.4 EMC test. The EMC requirements of 3.10 shall be verified by tests performed in accordance with MIL-STD-462.

4.5.5 Workmanship.

4.5.5.1 General workmanship. The control units, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirements of 3.13.1.

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4.5.5.2.1 Vibration screen. Prior to conducting temperature cycling, vibration shall be performed on each control unit. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All control units shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 1. The control accelerometer shall be located next to one of the mounting points of the control unit under vibration. Control units having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD). The control unit shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

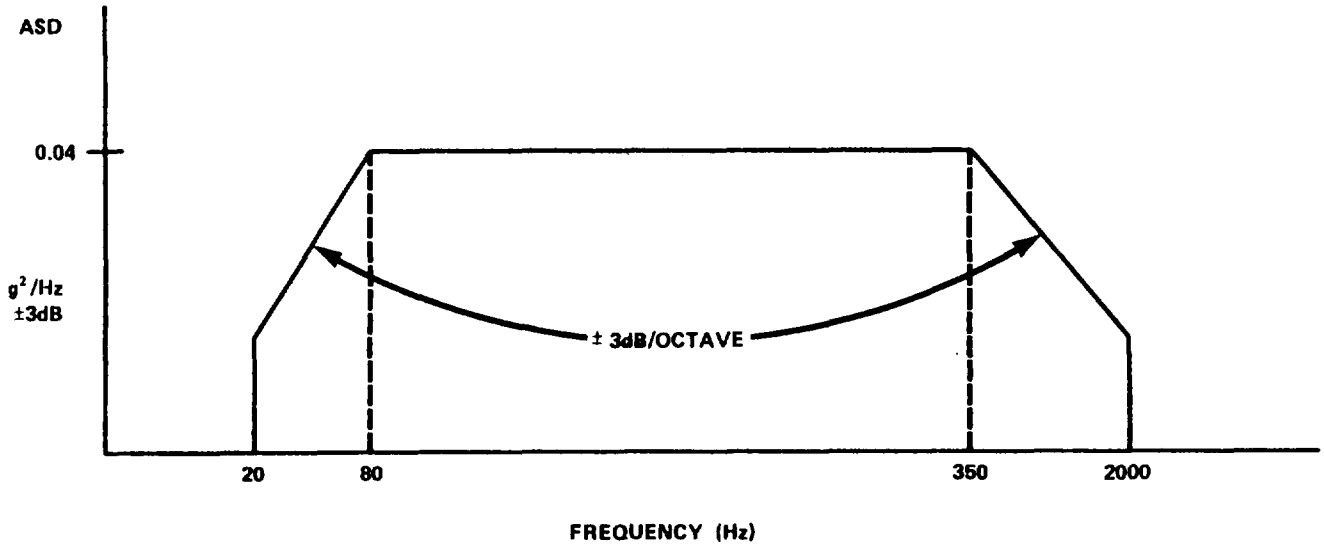


FIGURE 1. Random vibration curve.

4.5.5.2.2 Temperature cycling. Each control unit shall be subjected to 10 cycles of the temperature curve shown in FIGURE 2. The temperature rate of change shall not be less than 5°C per minute. Control unit power shall be turned on and off at the indicated times. The control unit shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the control unit reaches 80 percent of the chamber temperature. When failures occur, the unit shall be reworked and the cycling continued for a cummulative total of 10 cycles.

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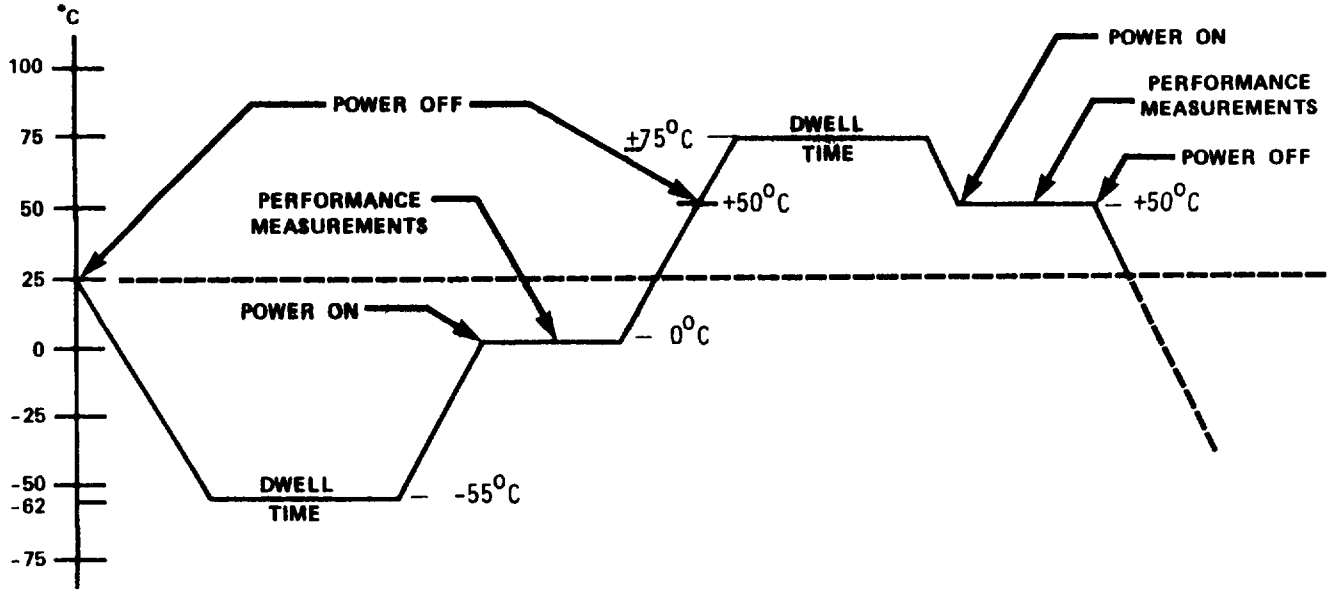


FIGURE 2. Temperature curve (one cycle).

4.5.6 Leakage current test.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE CONTROL UNIT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

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4.5.6.1 Control unit test connections. After power removal, each control unit directly connected to an external power source and units deriving power from the control unit shall be placed on an insulated surface. All safety ground conductors between the control units and units deriving power from the control units shall be intact. The safety ground conductor between the control units and the source power shall be opened during the test. The control unit shall be connected as shown in FIGURE 3 if it is connected to single-phase source power.

4.5.6.2 Measurement. Leakage current shall be measured on the control unit in its normal operating configuration. Control unit controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 volts at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the control unit is designed. The overall measurement error shall not exceed five percent. The probes shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed. The voltage shall be measured from each part to ground and from each part to all other simultaneously accessible parts for every combination of switch positions available in FIGURE 3. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.7 Crosstalk. The control units in operation shall be measured for minimum isolation for crosstalk in compliance with 3.8.7.

4.6 Reliability test. Production reliability acceptance testing shall be performed in accordance with Test Plan VC of MIL-STD-781 on a sample of at least three Type I and Type II control units randomly selected from the first month's production. The reliability test shall be successfully completed prior to delivery of any production control units under the contract.

4.6.1 Reliability test environment. Environmental conditions during the reliability test shall be as follows:

a. The ambient temperature shall be cycled between 0°C (+2°C, -0°C) for 3 hours and 50°C (+0°C, -2°C) for 3 hours.
b. Four Hz to 33 Hz to 4 Hz swept-sine vibration shall be applied in a single axis with vibration amplitude of 0.508 mm (0.020 in) (double amplitude (DA)) ±0.102 mm (±0.004 in) DA. The duration of each vibration sweep shall be 10 minutes ±2 minutes. The vibration shall be applied continuously for 1 hour every 3 hours.
c. Input power shall be changed daily in the following sequence:
    1. Nominal. 115 volts alternating current (VAC)
    2. Nominal. +7 percent
    3. Nominal. -7 percent

4.6.2 Operating duty cycle. All circuit elements shall be activated throughout the reliability test. Audio input signals and output impedance loads shall be used to simulate interfering equipments.

4.6.3 Failure. The procuring activity shall be notified within 48 hours of any failure during reliability testing. Failures will be classified by the procuring activity in accordance with MIL-STD-781.

4.6.4 Rejected control units. In the event of a reject decision in reliability testing, the contractor shall determine the cause(s) of failure and propose a plan for correction of deficiencies. After all proposed corrective actions have been approved by the procuring activity and incorporated in test samples, the reliability test shall be reperformed as specified herein.

4.6.5 Disposition of tested control units. Upon successful completion of reliability testing, the tested control unit shall be returned to the production area for reinspection and reconditioning as required and shall be subjected to the production tests of 4.4.1 to assure that it conforms to the requirements of this specification. After reconditioning, reinspection and retesting, the control unit may be delivered as part of the contract quantity, provided it conforms to all terms and conditions of the contract for acceptance.

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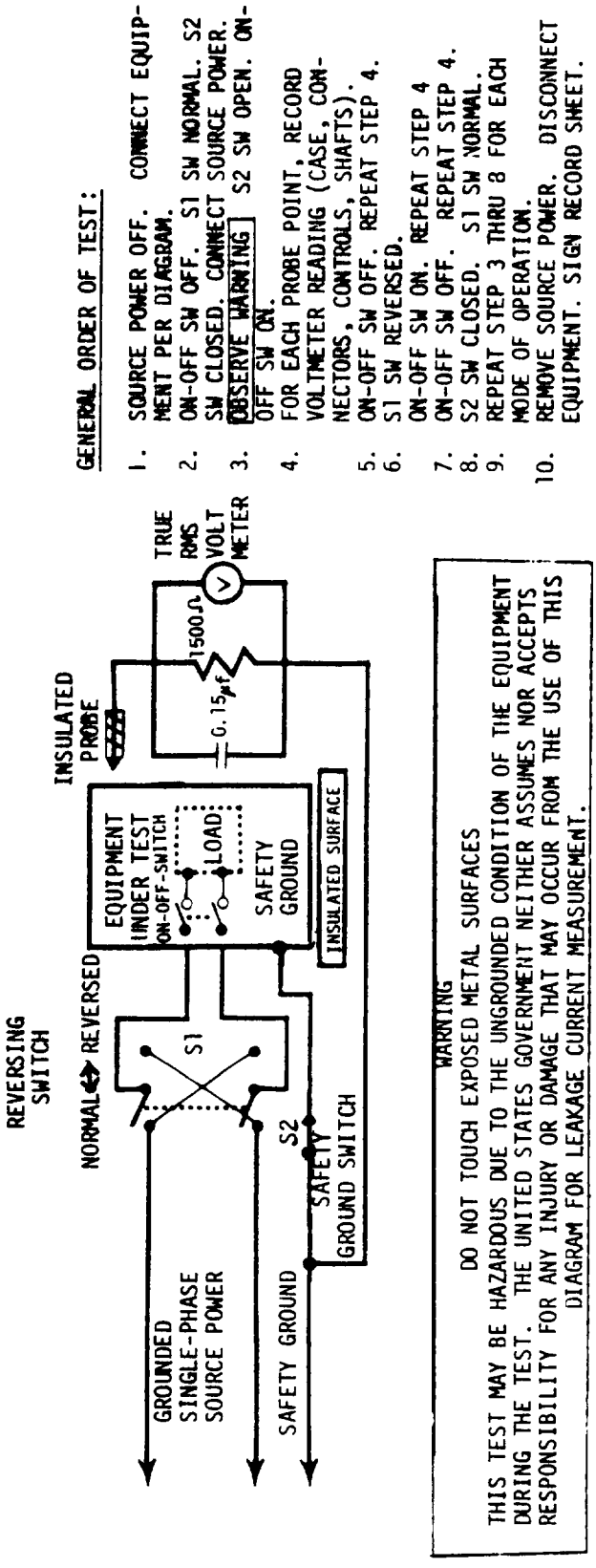


FIGURE 3. Single-phase test diagram for leakage current measurement.

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4.7 Maintainability demonstration. Conformance to the requirements of 3.7 shall be demonstrated in accordance with Method 4 of MIL-STD-471. Only Phase II of MIL-STD-471 shall apply. The term equipment-repair-time (ERT) used in MIL-STD-471 corresponds to M_{ct}. The demonstration shall consist of 20 corrective maintenance tasks for each Type I and Type II control unit requiring replacement of a subassembly or chassis-mounted part.

4.7.1 Selection of corrective maintenance tasks. For the maintainability demonstration, 40 candidate faults for each Type I and Type II control unit shall be determined in accordance with Appendix A of MIL-STD-471. The following information shall be made available for each candidate task:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part of simulation thereof)

The procuring activity or its authorized representative shall use the candidate tasks as a guide to select a sample of 20 faults for each Type I and Type II control unit for the demonstration.

4.7.2 Technical documentation. Technical documentation to be used for the demonstration shall be limited to the technical manual.

4.7.3 Accept or reject criteria. Accept or reject criteria shall be in accordance with Method 4 of MIL-STD-471. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies, and
b. Upon approval of an approach to the redesign or corrections, the contractor shall implement the approach and repeat the demonstration until an accept decision is reached.

4.8 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance to the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The Type I control unit covered by this specification is intended for use on surface ships and the Type II control unit is intended for use on small craft and in submarines. The control unit will be used in a secure voice communications system. It will provide an operator the capability of programming several remote telephone sets to secure radio channels.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Type required (see 1.2)
c. Number of first article samples to be submitted if other than specified in 4.3
d. Levels of preservation, packaging, packing, and marking (see 5.1)

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6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
3.4	Program Parts Selection List (PPSL)	DI-E-7027
3.4	Nonstandard Part Approval/Requests Proposed Additions To An Approved PPSL	DI-E-7028
4.3	First Article Inspection Report	DI-T-4902

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY-EC
(Project 5840-N203)

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	☐ VENDOR
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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